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F5-TTS: A Fairytaler that Fakes Fluent and Faithful Speech with Flow Matching

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Abstract

This paper introduces F5-TTS, a fully non-autoregressive text-to-speech system based on flow matching with Diffusion Transformer (DiT). Without requiring complex designs such as duration model, text encoder, and phoneme alignment, the text input is simply padded with filler tokens to the same length as input speech, and then the denoising is performed for speech generation, which was originally proved feasible by E2 TTS. However, the original design of E2 TTS makes it hard to follow due to its slow convergence and low robustness. To address these issues, we first model the input with ConvNeXt to refine the text representation, making it easy to align with the speech. We further propose an inference-time Sway Sampling strategy, which significantly improves our model's performance and efficiency. This sampling strategy for flow step can be easily applied to existing flow matching based models without retraining. Our design allows faster training and achieves an inference RTF of 0.15, which is greatly improved compared to state-of-the-art diffusion-based TTS models. Trained on a public 100K hours multilingual dataset, our Fairytaler Fakes Fluent and Faithful speech with Flow matching (F5-TTS) exhibits highly natural and expressive zero-shot ability, seamless code-switching capability, and speed control efficiency. Demo samples can be found at https://SWivid.github.io/F5-TTS. We release all code and checkpoints to promote community development[2].

1 Introduction

Recent research in Text-to-Speech (TTS) has experienced great advancement [1, 2, 3, 4, 5, 6, 7, 8]. With a few seconds of audio prompt, current TTS models are able to synthesize speech for any given text and mimic the speaker of audio prompt [9, 10]. The synthesized speech can achieve high fidelity and naturalness that they are almost indistinguishable from human speech [11, 12, 13, 14].

While autoregressive (AR) based TTS models exhibit an intuitive way of consecutively predicting the next token(s) and have achieved promising zero-shot TTS capability, the inherent limitations of AR modeling require extra efforts addressing issues such as inference latency and exposure bias [15, 16, 17, 18, 19]. Moreover, the quality of speech tokenizer is essential for AR models to achieve high-fidelity synthesis [20, 21, 22, 23, 24, 25, 26]. Thus, there have been studies exploring direct modeling in continuous space [27, 28, 29] to enhance synthesized speech quality recently.

Although AR models demonstrate impressive zero-shot performance as they perform implicit duration modeling and can leverage diverse sampling strategies, non-autoregressive (NAR) models

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[2]https://github.com/SWivid/F5-TTS

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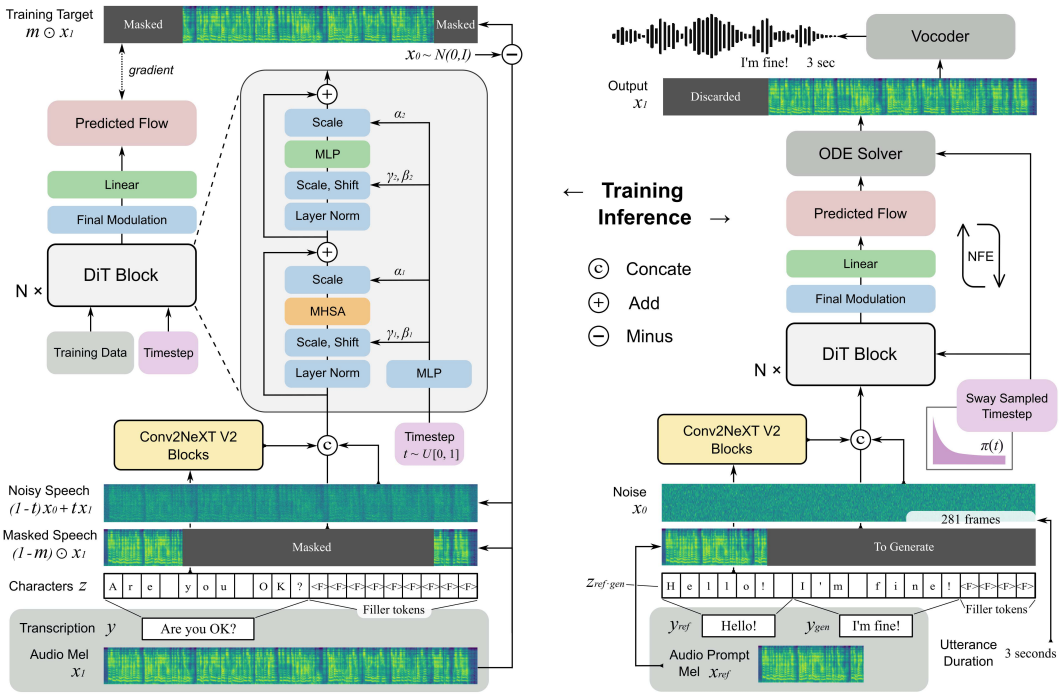

Figure 1: An overview of F5-TTS training (left) and inference (right). The model is trained on the text-guided speech-infilling task and condition flow matching loss. The input text is converted to a character sequence, padded with filler tokens to the same length as input speech, and refined by ConvNeXt blocks before concatenation with speech input. The inference leverages Sway Sampling for flow steps, with the model and an ODE solver to generate speech from sampled noise.

benefit from fast inference through parallel processing, and effectively balance synthesis quality and latency. Notably, diffusion models [30, 31] contribute most to the success of current NAR speech models [11, 12]. In particular, Flow Matching with Optimal Transport path (FM-OT) [32] is widely used in recent research fields not only text-to-speech [14, 33, 34, 35, 36] but also image generation [37] and music generation [38].

Unlike AR-based models, the alignment modeling between input text and synthesized speech is crucial and challenging for NAR-based models. While NaturalSpeech 3 [12] and Voicebox [14] use frame-wise phoneme alignment; Matcha-TTS [34] adopts monotonic alignment search and relies on phoneme-level duration model; recent works find that introducing such rigid and inflexible alignment between text and speech hinders the model from generating results with higher naturalness [36, 39].

E3 TTS [40] abandons phoneme-level duration and applies cross-attention on the input sequence but yields limited audio quality. DiTTo-TTS [35] uses Diffusion Transformer (DiT) [41] with cross-attention conditioned on encoded text from a pretrained language model. To further enhance alignment, it uses the pretrained language model to finetune the neural audio codec, infusing semantic information into the generated representations. In contrast, E2 TTS [36], based on Voicebox [14], adopts a simpler way, which removes the phoneme and duration predictor and directly uses characters padded with filler tokens to the length of mel spectrograms as input. This simple scheme also achieves very natural and realistic synthesized results. However, we found that robustness issues exist in E2 TTS for the text and speech alignment. Seed-TTS [39] employs a similar strategy and achieves excellent results, though not elaborated in model details. In these ways of not explicitly modeling phoneme-level duration, models learn to assign the length of each word or phoneme according to the given total sequence length, resulting in improved prosody and rhythm.

In this paper, we propose **F5-TTS**, a **F**airytaler that **F**akes **F**luent and **F**aithful speech with **F**low matching. Maintaining the simplicity of pipeline without phoneme alignment, duration predictor, text encoder, and semantically infused codec model, F5-TTS leverages the Diffusion Transformer

with ConvNeXt V2 [42] to better tackle text-speech alignment during in-context learning. We stress the deep entanglement of semantic and acoustic features in the E2 TTS model design, which has inherent problems and will pose alignment failure issues that could not simply be solved with re-ranking. With in-depth ablation studies, our proposed F5-TTS demonstrates stronger robustness, in generating more faithful speech to the text prompt, while maintaining comparable speaker similarity. Additionally, we introduce an inference-time sampling strategy for flow steps substantially improving naturalness, intelligibility, and speaker similarity of generation. This approach can be seamlessly integrated into existing flow matching based models without retraining.

2 Preliminaries

2.1 Flow Matching

The Flow Matching (FM) objective is to match a probability path p_t from a simple distribution p_0, *e.g.*, the standard normal distribution $p(x) = \mathcal{N}(x|0, I)$, to p_1 approximating the data distribution q. In short, the FM loss regresses the vector field u_t with a neural network v_t as

$$\mathcal{L}_{FM}(\theta) = E_{t,p_t(x)} \left\| v_t(x) - u_t(x) \right\|^2, \tag{1}$$

where θ parameterizes the neural network, $t \sim \mathcal{U}[0, 1]$ and $x \sim p_t(x)$. The model v_t is trained over the entire flow step and data range, ensuring it learns to handle the entire transformation process from the initial distribution to the target distribution.

利用训练数据构建条件概率路径 使其学到每个时间步t,每个位置x对应的目标移动方向

As we have no prior knowledge of how to approximate p_t and u_t, a conditional probability path $p_t(x|x_1) = \mathcal{N}(x \mid \mu_t(x_1), \sigma_t(x_1)^2 I)$ is considered in actual training, and the Conditional Flow Matching (CFM) loss is proved to have identical gradients *w.r.t.* θ [32]. x_1 is the random variable corresponding to training data. μ and σ is the time-dependent mean and scalar standard deviation of Gaussian distribution.

Remember that the goal is to construct target distribution (data samples) from initial simple distribution, *e.g.*, Gaussian noise. With the conditional form, the flow map $\psi_t(x) = \sigma_t(x_1)x + \mu_t(x_1)$ with $\mu_0(x_1) = 0$ and $\sigma_0(x_1) = 1$, $\mu_1(x_1) = x_1$ and $\sigma_1(x_1) = 0$ is made to have all conditional probability paths converging to p_0 and p_1 at the start and end. The flow thus provides a vector field $d\psi_t(x_0)/dt = u_t(\psi_t(x_0)|x_1)$. Reparameterize $p_t(x|x_1)$ with x_0, we have

$$\mathcal{L}_{\text{CFM}}(\theta) = E_{t,q(x_1),p(x_0)} \| v_t(\psi_t(x_0)) - \frac{d}{dt}\psi_t(x_0) \|^2. \tag{2}$$

Further leveraging Optimal Transport form $\psi_t(x) = (1 - t)x + tx_1$, we have the OT-CFM loss,

$$\mathcal{L}_{\text{CFM}}(\theta) = E_{t,q(x_1),p(x_0)} \| v_t((1 - t)x_0 + tx_1) - (x_1 - x_0) \|^2. \tag{3}$$

To view in a more general way [43], if formulating the loss in terms of log signal-to-noise ratio (log-SNR) λ instead of flow step t, and parameterizing to predict x_0 (ϵ, commonly stated in diffusion model) instead of predict $x_1 - x_0$, the CFM loss is equivalent to the v-prediction [44] loss with cosine schedule.

For inference, given sampled noise x_0 from initial distribution p_0, flow step $t \in [0, 1]$ and condition with respect to generation task, the ordinary differential equation (ODE) solver [45] is used to evaluate $\psi_1(x_0)$ the integration of $d\psi_t(x_0)/dt$ with $\psi_0(x_0) = x_0$. The number of function evaluations (NFE) is the times going through the neural network as we may provide multiple flow step values from 0 to 1 as input to approximate the integration. Higher NFE will produce more accurate results and certainly take more calculation time.

2.2 Classifier-Free Guidance

Classifier Guidance (CG) is proposed by [46], functions by adding the gradient of an additional classifier, while such an explicit way to condition the generation process may have several problems. Extra training of the classifier is required and the generation result is directly affected by the quality of the classifier. Adversarial attacks might also occur as the guidance is introduced through the way of updating the gradient. Thus deceptive images with imperceptible details to human eyes may be generated, which are not conditional.

Classifier-Free Guidance (CFG) [47] proposes to replace the explicit classifier with an implicit classifier without directly computing the explicit classifier and its gradient. The gradient of a classifier can be expressed as a combination of conditional generation probability and unconditional generation probability. By dropping the condition with a certain rate during training, and linear extrapolating the inference outputs with and without condition c, the final guided result is obtained. We could balance between fidelity and diversity of the generated samples with

$$v_{t,CFG} = v_t(\psi_t(x_0), c) + \alpha(v_t(\psi_t(x_0), c) - v_t(\psi_t(x_0))) \tag{4}$$

in CFM case, where α is the CFG strength.[3]

3 Method

This work aims to build a high-level text-to-speech synthesis system. We trained our model on the text-guided speech-infilling task [48, 14]. Based on recent research [35, 36, 49], it is promising to train without phoneme-level duration predictor and can achieve higher naturalness in zero-shot generation deprecating explicit phoneme-level alignment. We adopt a similar pipeline as E2 TTS [36] and propose our advanced architecture F5-TTS, addressing the slow convergence (timbre learned well at an early stage but struggled to learn alignment) and robustness issues (failures on hard case generation) of E2 TTS. We also propose a Sway Sampling strategy for flow steps at inference, which significantly improves our model's performance in faithfulness to reference text and speaker similarity.

3.1 Pipeline

Training The infilling task is to predict a segment of speech given its surrounding audio and full text (for both surrounding transcription and the part to generate). For simplicity, we reuse the symbol x to denote an audio sample and y the corresponding transcript for a data pair (x, y). As shown in Fig.1 (left), the acoustic input for training is an extracted mel spectrogram features $x_1 \in \mathbb{R}^{F \times N}$ from the audio sample x, where F is mel dimension and N is the sequence length. In the scope of CFM, we pass in the model the noisy speech $(1-t)x_0 + tx_1$ and the masked speech $(1-m) \odot x_1$, where x_0 denotes sampled Gaussian noise, t is sampled flow step, and $m \in \{0, 1\}^{F \times N}$ represents a binary temporal mask.

Following E2 TTS, we directly use alphabets and symbols for English. We opt for full pinyin to facilitate Chinese zero-shot generation. By breaking the raw text into such character sequence and padding it with filler tokens to the same length as mel frames, we form an extended sequence z with c_i denoting the i-th character:

$$z = (c_1, c_2, \ldots, c_M, \underbrace{\langle F \rangle, \ldots, \langle F \rangle}_{(N-M) \text{ times}}). \tag{5}$$

The model is trained to reconstruct $m \odot x_1$ with $(1-m) \odot x_1$ and z, which equals to learn the target distribution p_1 in form of $P(m \odot x_1 | (1-m) \odot x_1, z)$ approximating real data distribution q.

Inference To generate a speech with the desired content, we have the audio prompt's mel spectrogram features x_{ref}, its transcription y_{ref}, and a text prompt y_{gen}. Audio prompt serves to provide speaker characteristics and text prompt is to guide the content of generated speech.

The sequence length N, or duration, has now become a pivotal factor that necessitates informing the model of the desired length for sample generation. One could train a separate model to predict and deliver the duration based on x_{ref}, y_{ref} and y_{gen}. Here we simply estimate the duration based on the ratio of the number of characters in y_{gen} and y_{ref}. We assume that the sum-up length of characters is no longer than mel length, thus padding with filler tokens is done as during training.

To sample from the learned distribution, the converted mel features x_{ref}, along with concatenated and extended character sequence $z_{ref \cdot gen}$ serve as the condition in Eq.4. We have

$$v_t(\psi_t(x_0), c) = v_t((1-t)x_0 + tx_1 | x_{ref}, z_{ref \cdot gen}), \tag{6}$$

[3]Note that the inference time will be doubled if CFG. Model v_t will execute the forward process twice, once with condition, and once without.

See from Fig.1 (right), we start from a sampled noise x_0, and what we want is the other end of flow x_1. Thus we use the ODE solver to gradually integrate from $\psi_0(x_0) = x_0$ to $\psi_1(x_0) = x_1$, given $d\psi_t(x_0)/dt = v_t(\psi_t(x_0), x_{ref}, z_{ref \cdot gen})$. During inference, the flow steps are provided in an ordered way, $e.g.$, uniformly sampled a certain number from 0 to 1 according to the NFE setting.

After getting the generated mel with model v_t and ODE solver, we discard the part of x_{ref}. Then we leverage a vocoder to convert the mel back to speech signal.

3.2 F5-TTS

E2 TTS directly concatenates the padded character sequence with input speech, thus deeply entangling semantic and acoustic features with a large length gap of effective information, which is the underlying cause of hard training and poses several problems in a zero-shot scenario (Sec.5.1). To alleviate the problem of slow convergence and low robustness, we propose F5-TTS which accelerates training and inference and shows a strong robustness in generation. Also, an inference-time Sway Sampling is introduced, which allows inference faster (using less NFE) while maintaining performance. This sampling way of flow step can be directly applied to other CFM models.

Model As shown in Fig.1, we use latent Diffusion Transformer (DiT) [41] as backbone. To be specific, we use DiT blocks with zero-initialized adaptive Layer Norm (adaLN-zero). To enhance the model's alignment ability, we also leverage ConvNeXt V2 blocks [42]. Its predecessor ConvNeXt V1 [50] is used in many works and shows a strong temporal modeling capability in speech domain tasks [51, 52].

As described in Sec.3.1, the model input is character sequence, noisy speech, and masked speech. Before concatenation in the feature dimension, the character sequence first goes through ConvNeXt blocks. Experiments have shown that this way of providing individual modeling space allows text input to better prepare itself before later in-context learning. Unlike the phoneme-level force alignment done in Voicebox, a rigid boundary for text is not explicitly introduced. The semantic and acoustic features are jointly learned with the entire model. And unlike the way of feeding the model with inputs of significant length difference (length with effective information) as E2 TTS does, our design mitigates such gap.

The flow step t for CFM is provided as the condition of adaLN-zero rather than appended to the concatenated input sequence in Voicebox. We found that an additional mean pooled token of text sequence for adaLN condition is not essential for the TTS task, maybe because the TTS task requires more rigorously guided results and the mean pooled text token is more coarse.

We adopt some position embedding settings in Voicebox. The flow step is embedded with a sinusoidal position. The concatenated input sequence is added with a convolutional position embedding. We apply a rotary position embedding (RoPE) [53] for self-attention rather than symmetric bi-directional ALiBi bias [54]. And for extended character sequence $\hat{y}$, we also add it with an absolute sinusoidal position embedding before feeding it into ConvNeXt blocks.

Compared with Voicebox and E2 TTS, we abandoned the U-Net [55] style skip connection structure and switched to using DiT with adaLN-zero. Without a phoneme-level duration predictor and explicit alignment process, and nor with extra text encoder and semantically infused neural codec model in DiTTo-TTS, we give the text input a little freedom (individual modeling space) to let it prepare itself before concatenation and in-context learning with speech input.

Sampling As stated in Sec.2.1, the CFM could be viewed as v-prediction with a cosine schedule. For image synthesis, [37] propose to further schedule the flow step with a single-peak logit-normal [56] sampling, in order to give more weight to intermediate flow steps by sampling them more frequently. We speculate that such sampling distributes the model's learning difficulty more evenly over different flow step $t \in [0, 1]$.

In contrast, we train our model with traditional uniformly sampled flow step $t \sim \mathcal{U}[0, 1]$ but apply a non-uniform sampling during inference. In specific, we define a **Sway Sampling** function as

$$f_{sway}(u; s) = u + s \cdot (\cos(\frac{\pi}{2}u) - 1 + u), \tag{7}$$

which is monotonic with coefficient $s \in [-1, \frac{2}{\pi - 2}]$. We first sample $u \sim \mathcal{U}[0, 1]$, then apply this function to obtain sway sampled flow step t. With $s < 0$, the sampling is sway to left; with $s > 0$, the

sampling is sway to right; and $s = 0$ case equals to uniform sampling. Fig.3 shows the probability density function of Sway Sampling on flow step t.

Conceptually, CFM models focus more on sketching the contours of speech in the early stage ($t \to 0$) from pure noise and later focus more on the embellishment of fine-grained details. Therefore, the alignment between speech and text will be determined based on the first few generated results. With a scale parameter $s < 0$, we make model inference more with smaller t, thus providing the ODE solver with more startup information to evaluate more precisely in initial integration steps.

4 Experimental Setup

Datasets We utilize the in-the-wild multilingual speech dataset Emilia [57] to train our base models. After simply filtering out transcription failure and misclassified language speech, we retain approximately 95K hours of English and Chinese data. We also trained small models for ablation study and architecture search on WenetSpeech4TTS [58] Premium subset, consisting of a 945 hours Mandarin corpus. Base model configurations are introduced below, and small model configurations are in Appendix B.1. Three test sets are adopted for evaluation, which are LibriSpeech-PC *test-clean* [59], Seed-TTS *test-en* [39] with 1088 samples from Common Voice [60], and Seed-TTS *test-zh* with 2020 samples from DiDiSpeech [61][4]. Most of the previous English-only models are evaluated on different subsets of LibriSpeech *test-clean* while the used prompt list is not released, which makes fair comparison difficult. Thus we build and release a 4-to-10-second LibriSpeech-PC subset with 1127 samples to facilitate community comparisons.

Training Our base models are trained to 1.2M updates with a batch size of 307,200 audio frames (0.91 hours), for over one week on 8 NVIDIA A100 80G GPUs. The AdamW optimizer [62] is used with a peak learning rate of 7.5e-5, linearly warmed up for 20K updates, and linearly decays over the rest of the training. We set 1 for the max gradient norm clip. The F5-TTS base model has 22 layers, 16 attention heads, 1024/2048 embedding/feed-forward network (FFN) dimension for DiT; and 4 layers, 512/1024 embedding/FFN dimension for ConvNeXt V2; in total 335.8M parameters. The reproduced E2 TTS, a 333.2M flat U-Net equipped Transformer, has 24 layers, 16 attention heads, and 1024/4096 embedding/FFN dimension. Both models use RoPE as mentioned in Sec.3.2, a dropout rate of 0.1 for attention and FFN, the same convolutional position embedding as in Voicebox[14].

We directly use alphabets and symbols for English, use jieba[5] and pypinyin[6] to process raw Chinese characters to full pinyins. The character embedding vocabulary size is 2546, counting in the special filler token and all other language characters exist in the Emilia dataset as there are many code-switched sentences. For audio samples we use 100-dimensional log mel-filterbank features with 24 kHz sampling rate and hop length 256. A random 70% to 100% of mel frames is masked for infilling task training. For CFG (Sec.2.2) training, first the masked speech input is dropped with a rate of 0.3, then the masked speech again but with text input together is dropped with a rate of 0.2. We assume that the two-stage control of CFG training may have the model learn more with text alignment.

Inference The inference process is mainly elaborated in Sec.3.1. We use the Exponential Moving Averaged (EMA) [63] weights for inference, and the Euler ODE solver for F5-TTS (midpoint for E2 TTS as described in [36]). We use the pretrained vocoder Vocos [51] to convert generated log mel spectrograms to audio signals.

Baselines We compare our models with leading TTS systems including, (mainly) autoregressive models: VALL-E 2 [13], MELLE [29], FireRedTTS [64] and CosyVoice [65]; non-autoregressive models: Voicebox [14], NaturalSpeech 3 [12], DiTTo-TTS [35], MaskGCT [66], Seed-TTS$_{DiT}$ [39] and our reproduced E2 TTS [36]. Details of compared models see Appendix A.

Metrics We measure the performances under *cross-sentence* task. The model is given a reference text, a short speech prompt, and its transcription, and made to synthesize a speech reading the reference text mimicking the speech prompt speaker. In specific, we report Word Error Rate (WER) and speaker Similarity between generated and the original target speeches (SIM-o) for objective evaluation. For WER, we employ Whisper-large-v3 [67] to transcribe English and Paraformer-zh [68]

[4] `https://github.com/BytedanceSpeech/seed-tts-eval`

[5] `https://github.com/fxsjy/jieba`

[6] `https://github.com/mozillazg/python-pinyin`

Table 1: Results on LibriSpeech *test-clean* and LibriSpeech-PC *test-clean*. The boldface indicates the best result, and * denotes the score reported in baseline papers with different subsets for evaluation. The Real-Time Factor (RTF) is computed with the inference time of 10s speech. #Param. stands for the number of learnable parameters and #Data refers to the used training dataset in hours.

Model	#Param.	#Data(hrs)	WER(%)↓	SIM-o ↑	RTF ↓
LibriSpeech *test-clean*					
Ground Truth (*2.2 hours subset*)	-	-	2.2*	0.754*	-
VALL-E 2 [13]	-	50K EN	2.44*	0.643*	0.732*
MELLE [29]	-	50K EN	2.10*	0.625*	0.549*
MELLE-*R2* [29]	-	50K EN	2.14*	0.608*	0.276*
Voicebox [14]	330M	60K EN	**1.9***	**0.662***	0.64*
DiTTo-TTS [35]	740M	55K EN	2.56*	0.627*	**0.162***
Ground Truth (*40 samples subset*)	-	-	1.94*	0.68*	-
Voicebox [14]	330M	60K EN	2.03*	0.64*	0.64*
NaturalSpeech 3 [12]	500M	60K EN	**1.94***	0.67*	0.296*
MaskGCT [66]	1048M	100K Multi.	2.634*	**0.687***	-
LibriSpeech-PC *test-clean*					
Ground Truth (*1127 samples 2 hrs*)	-	-	2.23	0.69	-
Vocoder Resynthesized	-	-	2.32	0.66	-
CosyVoice [65]	∼300M	170K Multi.	3.59	0.66	0.92
FireRedTTS [64]	∼580M	248K Multi.	2.69	0.47	0.84
E2 TTS (32 NFE) [36]	333M	100K Multi.	2.95	**0.69**	0.68
F5-TTS (16 NFE)	336M	100K Multi.	2.53	0.66	**0.15**
F5-TTS (32 NFE)	336M	100K Multi.	**2.42**	0.66	0.31

for Chinese, following [39]. For SIM-o, we use a WavLM-large-based [69] speaker verification model to extract speaker embeddings for calculating the cosine similarity of synthesized and ground truth speeches. We use Comparative Mean Opinion Scores (CMOS) and Similarity Mean Opinion Scores (SMOS) for subjective evaluation. For CMOS, human evaluators are given randomly ordered synthesized speech and ground truth, and are to decide how higher the naturalness of the better one surpasses the counterpart, *w.r.t.* prompt speech. For SMOS, human evaluators are to score the similarity between the synthesized and prompt.

5 Experimental Results

Tab.1 and 2 show the main results of objective and subjective evaluations. We report the average score of three random seed generation results with our model and open-sourced baselines. We use by default a CFG strength of 2 and a Sway Sampling coefficient of -1 for our F5-TTS.

For English zero-shot evaluation, the previous works are hard to compare directly as they use different subsets of LibriSpeech *test-clean* [70]. Although most of them claim to filter out 4-to-10-second utterances as the generation target, the corresponding prompt audios used are not released. Therefore, we build a 4-to-10-second sample test set based on LibriSpeech-PC [59] which is an extension of LibriSpeech with additional punctuation marks and casing. To facilitate future comparison, we release the 2-hour test set with 1,127 samples, sourced from 39 speakers (LibriSpeech-PC missing one speaker).

F5-TTS achieves a WER of 2.42 on LibriSpeech-PC *test-clean* with 32 NFE and Sway Sampling, demonstrating its robustness in zero-shot generation. Inference with 16 NFE, F5-TTS gains an RTF of 0.15 while still supporting high-quality generation with a WER of 2.53. It is clear that the Sway Sampling strategy greatly improves performance. The reproduced E2 TTS shows an excellent speaker similarity (SIM) but much worse WER in the zero-shot scenario, indicating the inherent deficiency of alignment robustness.

Table 2: Results on two test sets, Seed-TTS *test-en* and *test-zh*. The boldface indicates the best result, the underline denotes the second best, and * denotes scores reported in baseline papers.

Model	**WER(%)**$\downarrow$	**SIM-o** $\uparrow$	**CMOS** $\uparrow$	**SMOS** $\uparrow$
		Seed-TTS *test-en*		
Ground Truth	2.06	0.73	0.00	3.91
Vocoder Resynthesized	2.09	0.70	-	-
CosyVoice [65]	3.39	0.64	0.02	3.64
FireRedTTS [64]	3.82	0.46	-1.46	2.94
MaskGCT [66]	2.623*	<u>0.717</u>*	-	-
Seed-TTS$_{DiT}$[39]	**1.733***	**0.790***	-	-
E2 TTS (32 NFE) [36]	2.19	0.71	0.06	<u>3.81</u>
F5-TTS (16 NFE)	1.89	0.67	<u>0.16</u>	3.79
F5-TTS (32 NFE)	<u>1.83</u>	0.67	**0.31**	**3.89**
		Seed-TTS *test-zh*		
Ground Truth	1.26	0.76	0.00	3.72
Vocoder Resynthesized	1.27	0.72	-	-
CosyVoice [65]	3.10	0.75	-0.06	3.54
FireRedTTS [64]	<u>1.51</u>	0.63	-0.49	3.28
MaskGCT [66]	2.273*	<u>0.774</u>*	-	-
Seed-TTS$_{DiT}$[39]	**1.178***	**0.809***	-	-
E2 TTS (32 NFE) [36]	1.97	0.73	-0.04	3.44
F5-TTS (16 NFE)	1.74	0.75	<u>0.02</u>	<u>3.72</u>
F5-TTS (32 NFE)	1.56	0.76	**0.21**	**3.83**

From the evaluation results on the Seed-TTS test sets, F5-TTS behaves similarly with a close WER to ground truth and comparable SIM scores. It produces smooth and fluent speech in zero-shot generation with a CMOS of 0.31 (0.21) and SMOS of 3.89 (3.83) on Seed-TTS *test-en* (*test-zh*), and surpasses some baseline models trained with larger scales. It is worth mentioning that Seed-TTS with the best result is trained with orders of larger model size and dataset (several million hours) than ours. As stated in Sec.3.1, we simply estimate duration based on the ratio of the audio prompt's transcript length and the text prompt length. If providing ground truth duration, F5-TTS with 32 NFE and Sway Sampling will have WER of 1.74 for *test-en* and 1.53 for *test-zh* while maintaining the same SIM, indicating a high upper bound. A robustness test on ELLA-V [15] hard sentences is further included in Appendix B.5.

5.1 Ablation of Model Architecture

To clarify our F5-TTS's efficiency and stress the limitation of E2 TTS. We conduct in-depth ablation studies. We trained small models to 800K updates (each on 8 NVIDIA RTX 3090 GPUs for one week), all scaled to around 155M parameters, on the WenetSpeech4TTS Premium 945 hours Mandarin dataset with half the batch size and the same optimizer and scheduler as base models. Details of small model configurations see Appendix B.1.

We first experiment with pure adaLN DiT (F5-TTS$-$*Conv2Text*), which fails to learn alignment given simply padded character sequences. Based on the concept of refining the input text representation to better align with speech modality, and keep the simplicity of system design, we propose to add jointly learned structure to the input context. Specifically, we leverage ConvNeXt's capabilities of capturing local connections, multi-scale features, and spatial invariance for the input text, which is our F5-TTS. And we ablate with adding the same branch for input speech, denoted F5-TTS$+$*Conv2Audio*. We further conduct experiments to figure out whether the long skip connection and the pre-refinement of input text are beneficial to the counterpart backbone, *i.e.* F5-TTS and E2 TTS, named F5-TTS$+$*LongSkip* and E2 TTS$+$*Conv2Text* respectively. We also tried with the Multi-Modal DiT (MMDiT) [37] a double-stream joint-attention structure for the TTS task which learned fast and collapsed fast, resulting in severe repeated utterance with wild timbre and prosody.

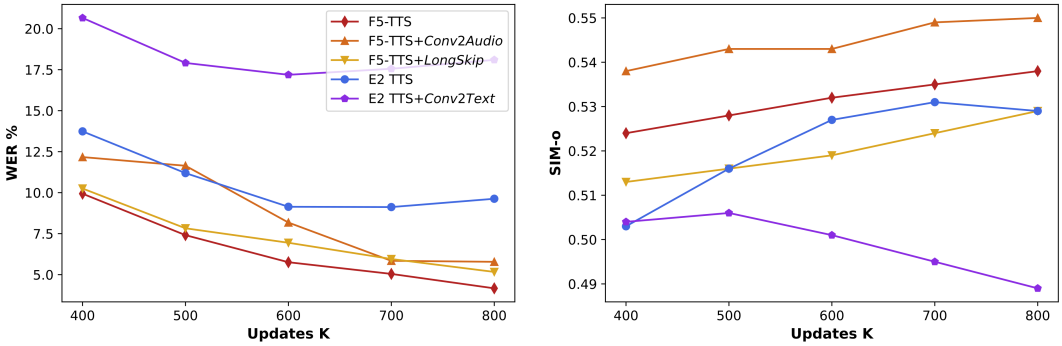

Figure 2: Ablation studies on model architecture. Seed-TTS *test-zh* evaluation results of 155M small models trained with WenetSpeech4TTS Premium a 945 hours Mandarin Corpus.

We assume that the pure MMDiT structure is far too flexible for rigorous task *e.g.* TTS which needs more faithful generation following the prompt guidance.

Fig.2 shows the overall trend of small models' WER and SIM scores evaluated on Seed-TTS *test-zh*. Trained with only 945 hours of data, F5-TTS (32 NFE *w/o* SS) achieves a WER of 4.17 and a SIM of 0.54 at 800K updates, while E2 TTS is 9.63 and 0.53. F5-TTS+*Conv2Audio* trades much alignment robustness (+1.61 WER) with a slightly higher speaker similarity (+0.01 SIM), which is not ideal for scaling up. We found that the long skip connection structure can not simply fit into DiT to improve speaker similarity, while the ConvNeXt for input text refinement can not directly apply to the flat U-Net Transformer to improve WER as well, both showing significant degradation of performance. To further analyze the unsatisfactory results with E2 TTS, we studied the consistent failure (unable to solve with re-ranking) on a 7% of the test set (WER>50%) all along the training process. We found that E2 TTS typically struggles with around 140 samples which we speculate to have a large distribution gap with the train set, while F5-TTS easily tackles this issue.

We investigate the models' behaviors with different input conditions to illustrate the advantages of F5-TTS further and disclose the possible reasons for E2 TTS's deficiency. See from Tab.4 in Appendix B.2, providing the ground truth duration allows more gains on WER for F5-TTS than E2 TTS, showing its robustness in alignment. By dropping the audio prompt, and synthesizing speech solely with the text prompt, E2 TTS is free of failures. This phenomenon implied a deep entanglement of semantic and acoustic features within E2 TTS's model design. From Tab.3 GFLOPs statistics, F5-TTS carries out faster training and inference than E2 TTS.

The aforementioned limitations of E2 TTS greatly hinder real-world application as the failed generation cannot be solved with re-ranking. Supervised fine-tuning facing out-of-domain data or a tremendous pretraining scale is mandatory for E2 TTS, which is inconvenient for industrial deployment. On the contrary, our F5-TTS better handles zero-shot generation, showing stronger robustness.

5.2 Ablation of Sway Sampling

It is clear from Fig.3 that a Sway Sampling with a negative s improves the generation results. Further with a more negative s, models achieve lower WER and higher SIM scores. We additionally include comparing results on base models with and without Sway Sampling in Appendix B.4.

As stated at the end of Sec.3.2, Sway Sampling with $s < 0$ scales more flow step toward early-stage inference ($t \to 0$), thus having CFM models capture more startup information to sketch the contours of target speech better. To be more concrete, we conduct a "leak and override" experiment. We first replace the Gaussian noise input x_0 at inference time with a ground-truth-information-leaked input $(1 - t')x_0 + t'x'_{ref}$, where $t' = 0.1$ and x'_{ref} is a duplicate of the audio prompt mel features. Then, we provide a text prompt different from the duplicated audio transcript and let the model continue the subsequent inference (skip the flow steps before t'). The model succeeds in overriding leaked utterances and producing speech following the text prompt if Sway Sampling is used, and fails without. Uniformly sampled flow steps will have the model producing speech dominated by

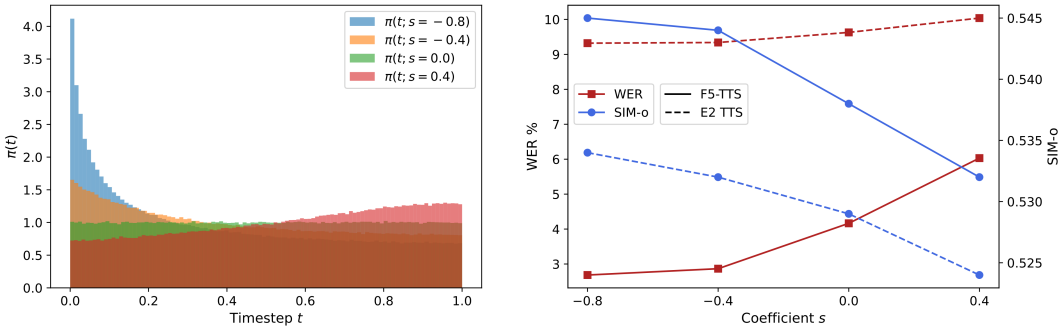

Figure 3: Probability density function of Sway Sampling on flow step t with different coefficient s (left), and small models' performance on Seed-TTS *test-zh* with Sway Sampling (right).

leaked information, speaking the duplicated audio prompt's context. Similarly, a leaked timbre can be overridden with another speaker's utterance as an audio prompt, leveraging Sway Sampling.

The experiment result is a shred of strong evidence proving that the early flow steps are crucial for sketching the silhouette of target speech based on given prompts faithfully, the later steps focus more on formed intermediate noisy output, where our sway-to-left sampling ($s < 0$) finds the profitable niche and takes advantage of it. We emphasize that our inference-time Sway Sampling can be easily applied to existing CFM-based models without retraining. And we will work in the future to combine it with training-time noise schedulers and distillation techniques to further boost efficiency.

6 Conclusion

This work introduces F5-TTS, a fully non-autoregressive text-to-speech system based on flow matching with diffusion transformer (DiT). With a tidy pipeline, literally text in and speech out, F5-TTS achieves state-of-the-art zero-shot ability compared to existing works trained on industry-scale data. We adopt ConvNeXt for text modeling and propose the test-time Sway Sampling strategy to further improve the robustness of speech generation and inference efficiency. Our design allows faster training and inference, by achieving a test-time RTF of 0.15, which is competitive with other heavily optimized TTS models of similar performance. We open-source our code, and models, to enhance transparency and facilitate reproducible research in this area.

Acknowledgments

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Ethics Statements

This work is purely a research project. F5-TTS is trained on large-scale public multilingual speech data and could synthesize speech of high naturalness and speaker similarity. Given the potential risks in the misuse of the model, such as spoofing voice identification, it should be imperative to implement watermarks and develop a detection model to identify audio outputs.

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A Baseline Details

VALL-E 2 [13] A large-scale TTS model shares the same architecture as VALL-E [9] but employs a repetition-aware sampling strategy that promotes more deliberate sampling choices, trained on Libriheavy [71] 50K hours English dataset. We compared with results reported in [29].

MELLE [29] An autoregressive large-scale model leverages continuous-valued tokens with variational inference for text-to-speech synthesis. Its variants allow to prediction of multiple mel-spectrogram frames at each time step, noted by MELLE-*Rx* with *x* denotes reduction factor. The model is trained on Libriheavy [71] 50K hours English dataset. We compared with results reported in [29].

Voicebox [14] A non-autoregressive large-scale model based on flow matching trained with infilling task. We compared with the 330M parameters trained on 60K hours dataset English-only model's results reported in [14] and [12].

NaturalSpeech 3 [12] A non-autoregressive large-scale TTS system leverages a factorized neural codec to decouple speech representations and a factorized diffusion model to generate speech based on disentangled attributes. The 500M base model is trained on Librilight [72] a 60K hours English dataset. We compared with scores reported in [12].

DiTTo-TTS [35] A large-scale non-autoregressive TTS model uses a cross-attention Diffusion Transformer and leverages a pretrained language model to enhance the alignment. We compare with DiTTo-en-XL, a 740M model trained on 55K hours English-only dataset, using scores reported in [35].

FireRedTTS [64] A foundation TTS framework for industry-level generative speech applications. The autoregressive text-to-semantic token model has 400M parameters and the token-to-waveform generation model has about half the parameters. The system is trained with 248K hours of labeled speech data. We use the official code and pre-trained checkpoint to evaluate[7].

MaskGCT [66] A large-scale non-autoregressive TTS model without precise alignment information between text and speech following the mask-and-predict learning paradigm. The model is multi-stage, with a 695M text-to-semantic model (T2S) and then a 353M semantic-to-acoustic (S2A) model. The model is trained on Emilia [57] dataset with around 100K Chinese and English in-the-wild speech data. We compare with results reported in [66].

Seed-TTS [39] A family of high-quality versatile speech generation models trained on unknown tremendously large data that is of orders of magnitudes larger than the previously largest TTS systems [39]. Seed-TTS$_{DiT}$ is a large-scale fully non-autoregressive model. We compare with results reported in [39].

E2 TTS [36] A fully non-autoregressive TTS system proposes to model without the phoneme-level alignment in Voicebox, originally trained on Libriheavy [71] 50K English dataset. We compare with our reproduced 333M multilingual E2 TTS trained on Emilia [57] dataset with around 100K Chinese and English in-the-wild speech data.

CosyVoice [65] A two-stage large-scale TTS system, first autoregressive text-to-token, then a flow matching diffusion model. The model is of around 300M parameters, trained on 170K hours of multilingual speech data. We obtain the evaluation result with the official code and pre-trained checkpoint[8].

[7] `https://github.com/FireRedTeam/FireRedTTS`

[8] `https://huggingface.co/model-scope/CosyVoice-300M`

B Experimental Result Supplements

B.1 Small Model Configuration

The detailed configuration of small models is shown in Tab.3. In the Transformer column, the numbers denote the Model Dimension, the Number of Layers, the Number of Heads, and the multiples of Hidden Size. In the ConvNeXt column, the numbers denote the Model Dimension, the Number of Layers, and the multiples of Hidden Size. GFLOPs are evaluated using the `thop` Python package.

As mentioned in Sec.3.2, F5-TTS leverages an adaLN DiT backbone, while E2 TTS is a flat U-Net equipped Transformer. F5-TTS+*LongSkip* adds an additional long skip structure connecting the first to the last layer in the Transformer. For the Multi-Model Diffusion Transformer (MMDiT) [37], a double stream transformer, the setting denotes one stream configuration.

Table 3: Details of small model configurations.

Model	**Transformer**	**ConvNeXt**	**#Param.**	**GFLOPs**
F5-TTS	768,18,12,2	512,4,2	158M	173
F5-TTS−*Conv2Text*	768,18,12,2	-	153M	164
F5-TTS+*Conv2Audio*	768,16,12,2	512,4,2	163M	181
F5-TTS+*LongSkip*	768,18,12,2	512,4,2	159M	175
E2 TTS	768,20,12,4	-	157M	293
E2 TTS+*Conv2Text*	768,20,12,4	512,4,2	161M	301
MMDiT [37]	512,16,16,2	-	151M	104

B.2 Ablation study on Input Condition

The ablation study on different input conditions is conducted with three settings: common input with text and audio prompts, providing ground truth duration information rather than an estimate, and retaining only text input dropping audio prompt. In Tab.4, all evaluations take the 155M small models' checkpoints trained on WenetSpeech4TTS Premium at 800K updates.

Table 4: Ablation study on different input conditions. The boldface indicates the best result, and the underline denotes the second best. All scores are the average of three random seed results.

Model	**Common Input**		**Ground Truth Dur.**		**Drop Audio Prompt**	
	WER ↓	**SIM ↑**	**WER ↓**	**SIM ↑**	**WER ↓**	**SIM ↑**
F5-TTS	**4.17**	<u>0.54</u>	**3.87**	<u>0.54</u>	<u>3.22</u>	0.21
F5-TTS+*Conv2Audio*	5.78	**0.55**	5.28	**0.55**	3.78	0.21
F5-TTS+*LongSkip*	<u>5.17</u>	0.53	<u>5.03</u>	0.53	3.35	0.21
E2 TTS	9.63	0.53	9.48	0.53	3.48	0.21
E2 TTS+*Conv2Text*	18.10	0.49	17.94	0.49	**3.06**	0.21

B.3 Comparison of ODE Solvers

The comparison results of using the Euler or midpoint ODE solver during F5-TTS inference are shown in Tab.5. The Euler is inherently faster (first-order) and performs slightly better typically for larger NFE inference with Sway Sampling (otherwise the Euler solver results in degradation).

B.4 Sway Sampling Effectiveness on Base Models

From Tab.6, it is clear that our Sway Sampling strategy for test-time flow steps consistently improves the zero-shot generation performance in aspects of faithfulness to prompt text (WER) and speaker similarity (SIM). The gain of applying Sway Sampling to E2 TTS [36] proves that our Sway Sampling strategy is universally applicable to existing flow matching based TTS models.

Table 5: Evaluation results of F5-TTS (F5) on LibriSpeech-PC *test-clean*, Seed-TTS *test-en* and Seed-TTS *test-zh*, employing the Euler or midpoint ODE solver, and with different Sway Sampling s values. The Real-Time Factor (RTF) is computed with the inference time of 10s speech.

Model	**LibriSpeech-PC *test-clean***		**Seed-TTS *test-en***		**Seed-TTS *test-zh***		
	WER(%)↓	**SIM-o ↑**	**WER↓**	**SIM-o ↑**	**WER↓**	**SIM-o ↑**	**RTF ↓**
Ground Truth	2.23	0.69	2.06	0.73	1.26	0.76	-
$s = -1$							
F5 (16 NFE Euler)	2.53	0.66	1.89	0.67	1.74	0.75	0.15
F5 (16 NFE midpoint)	2.43	0.66	1.88	0.66	1.61	0.75	0.26
F5 (32 NFE Euler)	2.42	0.66	1.83	0.67	1.56	0.76	0.31
F5 (32 NFE midpoint)	2.41	0.66	1.87	0.66	1.58	0.75	0.53
$s = -0.8$							
F5 (16 NFE Euler)	2.82	0.65	2.14	0.65	2.28	0.72	0.15
F5 (16 NFE midpoint)	2.58	0.65	1.86	0.65	1.70	0.73	0.26
F5 (32 NFE Euler)	2.50	0.66	1.81	0.67	1.62	0.75	0.31
F5 (32 NFE midpoint)	2.42	0.66	1.84	0.66	1.62	0.75	0.53

Table 6: Base model evaluation results on LibriSpeech-PC *test-clean*, Seed-TTS *test-en* and *test-zh*, with and without proposed test-time Sway Sampling (SS, with coefficient $s = -1$) strategy for flow steps. All generations leverage the midpoint ODE solver for ease of ablation.

Model	**WER(%)↓**	**SIM-o ↑**	**RTF ↓**
LibriSpeech-PC *test-clean*			
Ground Truth (*1127 samples*)	2.23	0.69	-
Vocoder Resynthesized	2.32	0.66	-
E2 TTS (16 NFE *w/* SS)	2.86	0.71	0.34
E2 TTS (32 NFE *w/* SS)	2.84	0.72	0.68
E2 TTS (32 NFE *w/o* SS)	2.95	0.69	0.68
F5-TTS (16 NFE *w/* SS)	2.43	0.66	0.26
F5-TTS (32 NFE *w/* SS)	2.41	0.66	0.53
F5-TTS (32 NFE *w/o* SS)	2.84	0.62	0.53
Seed-TTS *test-en*			
Ground Truth (*1088 samples*)	2.06	0.73	-
Vocoder Resynthesized	2.09	0.70	-
E2 TTS (16 NFE *w/* SS)	1.99	0.72	0.34
E2 TTS (32 NFE *w/* SS)	1.98	0.73	0.68
E2 TTS (32 NFE *w/o* SS)	2.19	0.71	0.68
F5-TTS (16 NFE *w/* SS)	1.88	0.66	0.26
F5-TTS (32 NFE *w/* SS)	1.87	0.66	0.53
F5-TTS (32 NFE *w/o* SS)	1.93	0.63	0.53
Seed-TTS *test-zh*			
Ground Truth (*2020 samples*)	1.26	0.76	-
Vocoder Resynthesized	1.27	0.72	-
E2 TTS (16 NFE *w/* SS)	1.80	0.78	0.34
E2 TTS (32 NFE *w/* SS)	1.77	0.78	0.68
E2 TTS (32 NFE *w/o* SS)	1.97	0.73	0.68
F5-TTS (16 NFE *w/* SS)	1.61	0.75	0.26
F5-TTS (32 NFE *w/* SS)	1.58	0.75	0.53
F5-TTS (32 NFE *w/o* SS)	1.93	0.69	0.53

B.5 ELLA-V Hard Sentences Evaluation

ELLA-V [15] proposed a challenging set containing 100 difficult textual patterns evaluating the robustness of the TTS model. Following previous works [13, 29, 36], we include generated samples in our demo page[9]. We additionally compare our model with the objective evaluation results reported in E1 TTS [49].

StyleTTS 2 is a TTS model leveraging style diffusion and adversarial training with large speech language models. CosyVoice is a two-stage large-scale TTS system, consisting of a text-to-token AR model and a token-to-speech flow matching model. Concurrent with our work, E1 TTS$_{DMD}$ is a diffusion-based NAR model with a distribution matching distillation technique to achieve one-step TTS generation. Since the prompts used by E1 TTS$_{DMD}$ are not released, we randomly sample 3-second-long speeches in our LibriSpeech-PC *test-clean* set as audio prompts. The evaluation result is in Tab.7. We evaluate the reproduced E2 TTS and our F5-TTS with 32 NFE and Sway Sampling and report the averaged score of three random seed results.

Table 7: Results of zero-shot TTS WER on ELLA-V hard sentences. The asterisk * denotes the score reported in E1 TTS. Sub. for substitution, Del. for Deletion, and Ins. for Insertion.

Model	**WER(%))↓**	**Sub.(%)↓**	**Del.(%)↓**	**Ins.(%)↓**
StyleTTS 2 [73]	4.83*	2.17*	2.03*	0.61*
CosyVoice [65]	8.30*	3.47*	2.74*	1.93*
E1 TTS$_{DMD}$ [49]	4.29*	1.89*	1.62*	0.74*
E2 TTS [36]	8.58	3.70	4.82	0.06
F5-TTS	4.40	1.81	2.40	0.18

We note that a higher WER compared to the results on commonly used test sets is partially due to mispronunciation (*yogis* to *yojus*, *cavorts* to *caverts*, *etc.*). The high Deletion rate indicates a word-skipping phenomenon when our model encounters a stack of repeating words. The low Insertion rate makes it clear that our model is free of endless repetition. We further emphasize that prompts from different speakers will spell very distinct utterances, where the ASR model transcribes correctly for one, and fails for another (*e.g. quokkas* to *Cocos*).

[9] `https://SWivid.github.io/F5-TTS`